

# The mass gap: current position of this programme

This is the synthesis of the notes written since the goal was set to a proof of the Yang–Mills existence and mass-gap conjecture. It states what has been proved here, what has been imported, what has been ruled out, and what remains, with constants explicit and each claim labelled.

The short version: the conjecture decomposes into six named statements, of which two are proved here and one is imported. The clause  $m < \infty$  is closed as far as variational methods reach, and reduced to the existence of the theory plus the nontriviality of one flowed correlator. The clause  $m > 0$  is untouched, and **seven routes to it have been closed by explicit computation**, in two families: those built from operator norms and spectra, and those built from the ground state of the Hamiltonian. Both families fail for reasons now stated exactly, and the second failure identifies what the Euclidean formulation supplies that the Hamiltonian one does not.

## 1. The decomposition

The obligations map turns “prove the mass gap” into six statements about the Kogut–Susskind Hamiltonian

$$H = \frac{\hbar c}{a} \left[ \frac{g^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g^2} \sum_p (N - \text{Re tr } U_p) \right], \quad \Delta_{a,L} = \frac{\hbar c}{a} \delta(g; N_s, G).$$

|     | statement                                                                                      | status                                                                                               |
|-----|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| T1  | finite lattice: unique physical ground state, $\delta > 0$                                     | <b>proved here</b>                                                                                   |
| T2  | $\delta \geq \gamma(g^2/2)C_2$ for $g \geq g_0$ , uniform in volume                            | <b>proved here</b> from an imported theorem; explicit for the Wilson transfer matrix, $g^2 \geq 176$ |
| T2' | $\liminf_{N_s} \delta > 0$ for every $g > 0$                                                   | open; false for $U(1)$                                                                               |
| T3  | $\delta_{\infty}(g)/(a\Lambda_{\text{lat}}(g)) \rightarrow m/(\hbar c\Lambda) \in (0, \infty)$ | open                                                                                                 |
| T4  | continuum infinite-volume theory with the axioms                                               | finite-volume ultraviolet stability known                                                            |
| S   | small volume:<br>$\Delta = \delta_1 g^{2/3} \hbar c / L [1 + O(g^{2/3})]$                      | upper side proved, lower side fixed-lattice only                                                     |

The conjecture is T2' together with T3, given T4, and the critical-coupling note shows that T2' is exactly the absence of a zero-temperature bulk phase transition, hence on its second-order branch the uniqueness of the continuum limit.

## 2. What is proved here

**T1** (obligations map §2): compact resolvent, a simple strictly positive ground state, a positive gap inherited by the physical sector, for every  $a$ ,  $N_s$ ,  $g$  and compact connected  $G$ .

**T2 in Hamiltonian form** (T2 note): the electric term is classical in the Peter–Weyl partition and the magnetic term is a bounded range- $\{0, 1\}^3$  perturbation with  $\beta = 48N^2/[g^4(N^2 - 1)]$ , so Yarotsky’s theorem gives  $\Delta_{a,L} \geq \gamma(g^2/2)C_2\hbar c/a$  for  $g \geq g_0$ , uniformly in the lattice size, with exponential clustering.

**A Lieb–Robinson bound** (note): velocity  $v \leq 32eNc/g^2$ , uniform in volume, because the unbounded part of  $H$  is a sum of commuting single-link Laplacians; hence a strong-coupling correlation length  $\xi \leq C'aN/(\gamma C_2 g^4)$ .

**Upper bounds** (note, Polyakov note): the Feynman–Bijl inequality with explicit constants; the strong-coupling gap pinned to order  $g^2\hbar c/a$  on both sides; the abelian gap bounded by the plaquette-sine structure factor; the proof that **no operator linear in the electric field with c-number coefficients is gauge invariant for a non-abelian group**; and a necessary condition on the Polyakov-loop susceptibility.

**The finiteness clause reduced** (moment hierarchy):  $m \leq M_{k+1}/M_k$  for every  $k$ , with the ratios decreasing, so the  $f$ -sum rule is the weakest member and the next one,  $m \leq \hbar C_0(0)/\int_0^\infty C_0$ , needs only T4 and  $C_0 \neq 0$ . The free-field value of the weakest member is  $4.26 \hbar c/\sqrt{8t}$ , computed in closed form and confirmed by an independent sum rule (free-field note).

**Exact structural identities** (note):  $\Delta V = 4C_2(N|\mathcal{P}| - V)$ , so the magnetic energy shifted by its Haar mean is a Laplacian eigenfunction;  $|\nabla V|^2 \leq 16V$  for  $SU(2)$ ; and an exact ground-state sum rule tying  $\text{Var}_\Omega(V)$  to kinetic quantities.

**Flow facts** (conjugation, lattice truncation): the Wilson flow is a diffeomorphism of  $G^\mathcal{E}$ , unitarily implemented, so conjugation preserves the spectrum; the lattice flow carries no coupling and a doubling is dimensionless flow time  $1/2$ , so truncating the conjugated Hamiltonian at range  $Ka$  costs  $Ce^{-cK}$  with pure constants.

### 3. Seven closed routes

| route                                 | why it closes                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------|
| variational upper bounds, for $m > 0$ | a spectral measure with all negative moments finite can have $m = 0$                          |
| expansion around the free theory      | the bound reaches $m$ only at the confinement scale, where its coefficient is nonperturbative |
| blocking by projection                | the boundary grows like $M^2$ while the block gap does not; the criterion worsens by $3M^2/8$ |
| conjugation by the flow               | the spectrum is invariant, so the gain is the truncation, which is cheap                      |
| flow before decimation                | sup norms are bijection-invariant and spectra conjugation-invariant                           |
| Agmon estimate on the ground state    | $\ \nabla V\ _\infty$ is extensive, so it bounds the global excess only                       |
| pointwise Gibbs domination            | the comparison function bites only above the extensive threshold $V_* \simeq N \mathcal{P} $  |

The first five exhaust the methods built from operator norms and spectra; the last two exhaust the standard ways of extracting a local statement from the ground-state eigenvalue equation. Each closure is a computation with explicit constants.

**The reason the last two fail, stated once.** Every inequality derived from  $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$  compares the total potential with the total energy, and  $e_0$  is extensive, so it separates configurations only by their global excess. The Euclidean weight  $e^{-S_w} = \prod_p e^{-(2/g_E^2)V_p}$  factorizes over plaquettes and separates them locally by construction. That is the structural advantage of the Euclidean formulation, derived here rather than assumed.

### 4. Where the non-abelian structure has been isolated

Three statements distinguish the groups, and any proof of T2' must use one of them.

1. *The commutator potential* of the zero-momentum sector is identically zero for an abelian group; where present, the transverse zero-point energy confines the valley and produces the gap  $\delta_1 \hbar^{4/3} g^{2/3} m^{-2/3}$  (G07, claim C133).

2. *Gauge invariance blocks the photon channel*: no c-number-coefficient operator linear in  $E$  commutes with Gauss's law for a non-abelian group, so the excitation through which the abelian gap closes must carry a Wilson line (upper-bound note Proposition 6).
3. *The one-loop valley potential*  $U(a) = \frac{2\hbar c}{\pi^2 L} \Phi(La)$  vanishes identically in the abelian theory, so the holonomy is free (valley note §3b).

## 5. What remains

| obligation                        | kind                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T2': closedness of the gapped set | the conjecture                                                                                                                                                                                                                                                                                              |
| openness of the gapped set        | technical: gap stability without frustration-freeness                                                                                                                                                                                                                                                       |
| T3                                | the conjecture, given T2'                                                                                                                                                                                                                                                                                   |
| T4                                | construction                                                                                                                                                                                                                                                                                                |
| local large-field control         | available via the transfer-matrix                                                                                                                                                                                                                                                                           |
|                                   | identification: $ \Omega ^2$ is the Euclidean time-slice marginal                                                                                                                                                                                                                                           |
| the large-field region itself     | closed in measure (cost $\eta^2/(4g^2)$ sharp, entropy $\eta^2/(8\pi^2)$ ); the Hamiltonian chain needs an operator inequality there, whose proof loses $C_V  \partial N  \hbar c/a$ on the small-field side, a negative perturbation that binds; the polymer expansion consumes measure statements instead |

The pattern across every note is uniform: **statements uniform in the cutoff are renormalization statements, and statements local in space need a local weight.** The strong-coupling theorem is uniform in the volume at fixed cutoff; the small-volume theorem is fixed-lattice; the flowed bounds are finite at fixed lattice and need the flow's renormalization in the continuum; the ground-state estimates are global because the eigenvalue equation is.

## 6. Assessment

The route that survives is the constructive one: conjugate by the flow, which is exact; truncate, which is cheap, inside the large-field region too once the range grows like  $\eta$ ; decimate, which is the problem; and iterate about twenty-one times for  $SU(3)$  from  $g_{UV}^2 = 1/2$  ( $SU(3)$  constants §3). The identification of  $\Omega^2$  with the Euclidean time-slice marginal (transfer note) imports every local Euclidean estimate, and the large-field region is then closed in measure: cost  $\eta^2/(4g^2)$ , sharp; entropy  $\eta^2/(8\pi^2)$ ; the flow's growth there is the Nielsen–Olesen mode and cannot be improved. What the Hamiltonian chain of lower bounds consumes from that region is an operator inequality, whose proof loses a negative term of order  $\hbar c/a$  on the small-field side, of exactly the kind that binds (operator-inequality note). The Euclidean polymer expansion consumes the measure statements directly, which is the reason the constructive programme is Euclidean, now stated as a computation.

**The coupling map for  $SU(3)$ .** Three regions, with numbers:

| region                                                                                    | statement                                                                                                                                                                   | source                                              |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| strong, $g^2 > 444$ (Dobrushin)<br>or $g^2 \geq 1056$ (polymer)                           | $\Delta_W \geq (\hbar c/a) \log(g^2/432)$ ,<br>resp. $4 \log(g^2/1056)$ , uniform<br>in volume, Wilson transfer<br>matrix                                                   | Dobrushin note, polymer note                        |
| strong, $g^2 \geq 388$ (79 in the<br>adjacent-growth class)<br>weak, $1/g^2 \gtrsim 10^2$ | $\Delta_{KS} \geq \frac{4}{3}g^2 \hbar c/a$ , approaching<br>$\frac{8}{3}g^2$ , Kogut–Susskind<br>small-field expansion applies,<br>crude window $C_1 g \leq \eta \leq C_2$ | continuous-time note<br>operator-inequality note §4 |
| intermediate,<br>$10^{-2} \lesssim g^2 \lesssim 4 \times 10^2$                            | no expansion applies; a<br>finite-volume mixing<br>condition<br>(Dobrushin–Shlosman) would<br>give the gap coupling by<br>coupling                                          | finite-verification note                            |

The strong boundary is explicit and the weak one is crude; the width of the intermediate region in one-loop doublings is set almost entirely by the weak side, about  $10^3$  with the present window and about 20 if the small-field expansion reaches  $g^2 \sim 1/2$  (threshold note §4). The Kogut–Susskind threshold from Yarotsky’s theorem, made explicit, is  $g_0^2 \sim 10^{101}$  and plays no role in the map.

**The map as a theorem.** The conditional theorem states the whole map as two hypotheses, control of the blocking steps from the weak side to the coupling where  $\xi \simeq a$  (H1) and certified mixing at that one coupling on one box (H2), with the conclusion  $m \geq \hbar c \min(\gamma_*, \gamma')/a_*$ , where  $a_* \simeq 0.1$  fm is the physical spacing at the verified coupling. Progress is progress on H1 or H2.

**Division of labour.** Hamiltonian methods for T1, T2, the small-volume theorem, the upper bounds, the exact identities and the final gap extraction; the Euclidean polymer expansion for the renormalization steps; and, for the intermediate region, the finite-volume mixing conditions, which reduce the gap at a given coupling to a finite verification and turn the problem into the entry of the trajectory of effective interactions into the open set of completely analytical interactions, a sufficient route whose two-number form is the meeting of the reach  $g_{RG}^2$  of the small-field renormalization with the reach  $g_{DS}^2$  of the verification (finite-verification note).

What this programme has added is a map with constants: six named statements, two proved, the finiteness clause reduced to a single correlator, seven routes closed with explicit numbers, the large-field region closed in measure and its obstruction in norm identified, three places where the non-abelian structure is isolated in a usable form, the exact identities and sum rule of Section 2, and explicit thresholds on both sides of the intermediate region. Several of the closures are corrections of claims made earlier in the same series, and each is recorded with the computation that forced it.